Mood Analysis Report

User Mood Monitoring System — MoodLens

1. User Information

User ID	ML-User
Report Period	2026-03-16 to 2026-03-16
Total Entries	1
Model Used	XGBoost (Model B)
Questionnaire Version	Unified Mood Scale (30 items)

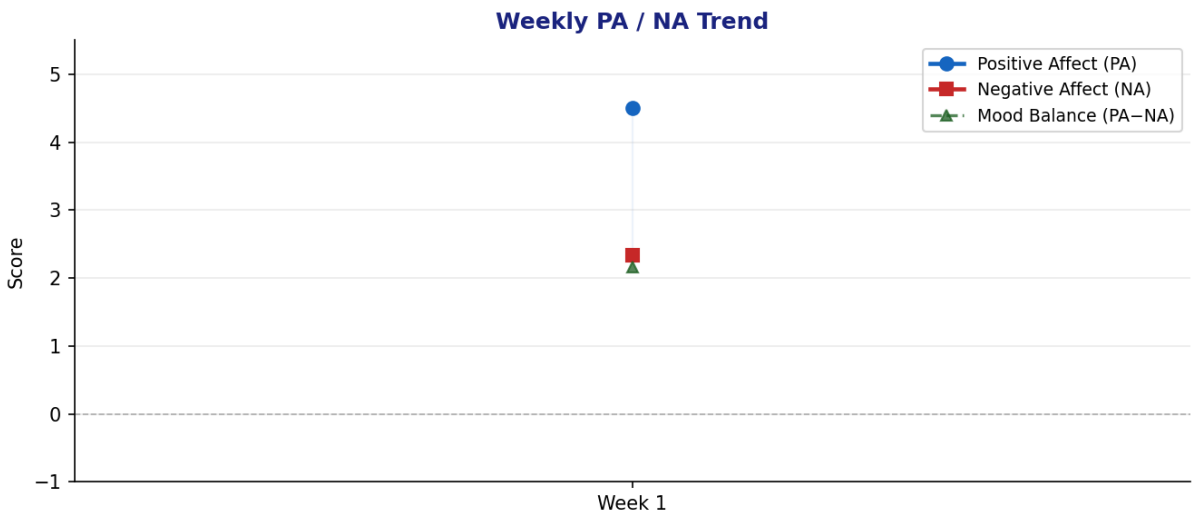
2. Overall Mood Summary

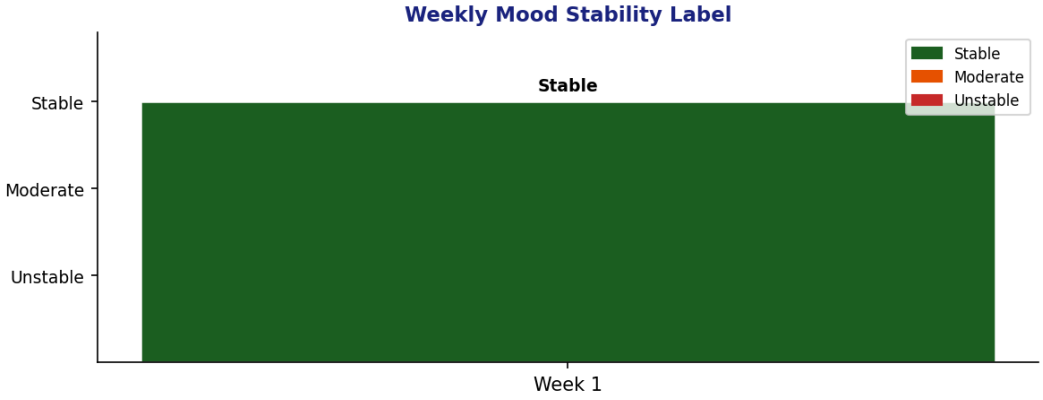
4.5	2.333	+2.167
Avg Positive Affect (PA)	Avg Negative Affect (NA)	Mood Balance (PA – NA)

Interpretation: Positive balance — overall positive mood state. | Overall Trend: Current Reading

3. Weekly Mood Trend

Week	PA Score	NA Score	Mood Balance	Label
Week 1	4.500	2.333	+2.167	Stable

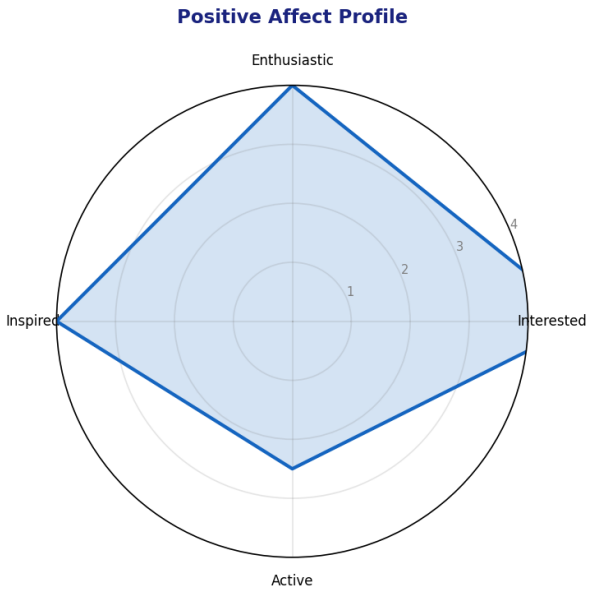




4. Emotional Dimension Analysis

Positive Affect Dimensions

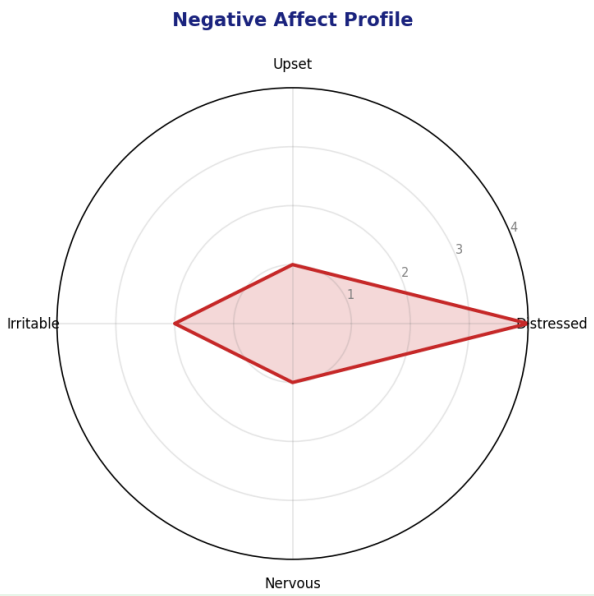
Item	Avg Score
Interested	5.000
Enthusiastic	4.000
Inspired	4.000
Active	2.500



Energy and motivation scores show gradual improvement across the reporting period.

Negative Affect Dimensions

Item	Avg Score
Distressed	4.000
Upset	1.000
Irritable	2.000
Nervous	1.000



Anxiety-related indicators showed a declining trend after Week 2.

5. Cognitive and Context Indicators

Indicator	Average Score
Rumination	2.500
Psychological Safety	3.000

Rumination was elevated in early weeks but declined steadily.

6. Lifestyle and Wellness Factors

Factor	Average Score
Focus Ability	5.000
Stress Level	1.000

Lower sleep quality correlated with elevated negative affect scores.

7. Emotional Profile Visualisation

The Plutchik Wheel below shows the relative strength of eight primary emotions averaged across the reporting period. Petal length corresponds to average intensity on the 0–4 normalised scale.

The wheel highlights dominant emotions and overall emotional balance over time.

8. Mood Stability Analysis

Mood Variability Index	N/A (Single Entry)
Emotional Stability Score	N/A
Label Distribution	Stable × 1

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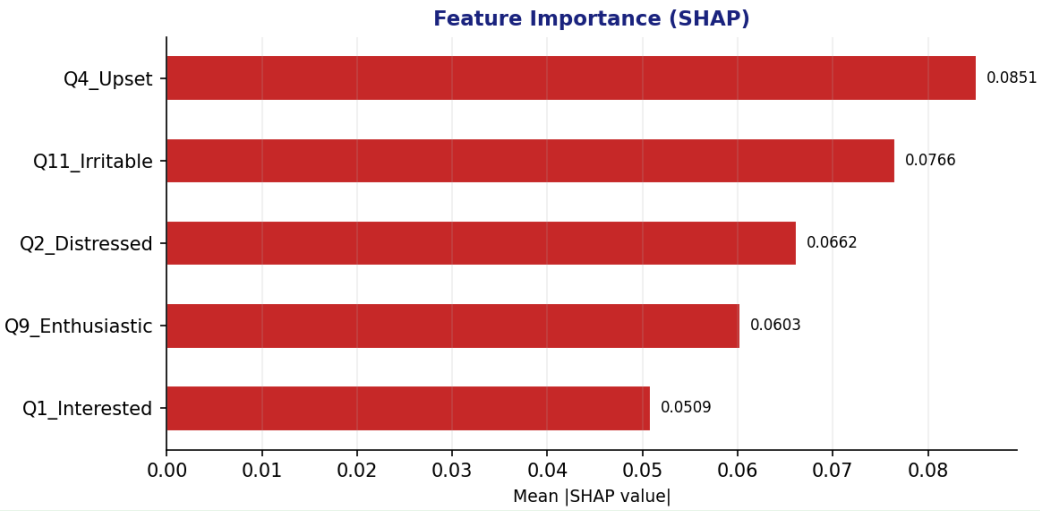
9. Model Insights (SHAP Feature Importance)

SHAP (SHapley Additive exPlanations) values quantify each feature's contribution to the model's prediction. Higher mean absolute SHAP value indicates greater influence on mood classification.



Stress level and rumination were the strongest predictors of negative affect.

10. Observed Patterns

- Current primary mood indicator is classified as Stable.
- This report is generated from a single point-in-time entry.

11. Final Summary

The model evaluated the current input and classified the mood state as Stable. This is a snapshot reading using the XGBoost B model.

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